

Homogeneous Linear Differential Equations with Constant Coefficients

Consider the following partial differential equations.

$$A \frac{\partial z}{\partial x} + B \frac{\partial z}{\partial y} = 0$$

$$A \frac{\partial^2 z}{\partial x^2} + B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} = 0$$

$$A \frac{\partial^2 z}{\partial x^2} + B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} = x + 2y + .$$

Here A, B, C are real constants.

Substitute $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$

Then, $(A D + B D')z = 0$

$$(A D^2 + B D D' + C D'^2)z = 0$$

$$(A D^2 + B D D' + C D'^2)z = x + 2y + 1$$

General Form

$$\left(A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + A_2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \cdots + A_n \frac{\partial^n z}{\partial y^n} \right) +$$
$$\left(B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + \cdots + B_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}} \right) + \cdots + \left(M_0 \frac{\partial z}{\partial x} + M_1 \frac{\partial z}{\partial y} \right) + N_0 z = f(x, y)$$

$A_0, \dots, A_n, B_0, \dots, B_{n-1}, M_0, M_1, N_0$ are constant functions.

$$F(D, D') z = f(x, y)$$

The general solution of partial differential equations comprises of two parts similar to that of ordinary differential equations. They are the complementary function and the particular integral.

Partial differential equation of the form

$$(A_0 D^n + A_1 D^{n-1} D'^{n-1} + \dots + A_n D'^n) z = f(x, y)$$

is a nonhomogeneous partial differential equation with constant coefficients.

Let $z = \phi(y + mx)$ is a solution

$$Dz = \frac{\partial z}{\partial x} = m \phi'(y + mx)$$

$$D^2 z = \frac{\partial^2 z}{\partial x^2} = m^2 \phi''(y + mx)$$

$$\vdots \qquad \qquad \vdots \qquad \qquad \vdots$$

$$D^n z = \frac{\partial^n z}{\partial x^n} = m^n \phi^{(n)}(y + mx)$$

$$D'z = \frac{\partial z}{\partial y} = \phi'(y + mx)$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$D'^n z = \frac{\partial^n z}{\partial y^n} = \phi^n(y + mx)$$

$$F(D, D')z = (A_0 m^n + A_1 m^{n-1} + \dots + A_{n-1} m + A_n) \phi^n(y + mx) = 0$$

$$A_0 m^n + A_1 m^{n-1} + \dots + A_{n-1} m + A_n = 0$$

This equation is called the Auxiliary equation.

Let $m_1, m_2, \dots, m_n$ be the n distinct roots of the above equation. Then the complementary function is of the form,

$$z = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x)$$

Working rule of finding **complementary function**

$$D' = 1$$

$$(A_0 D^n + A_1 D^{n-1} + \dots + A_n) = 0$$

Case I

When $D = m = \frac{a_1}{b_1}, \frac{a_2}{b_2}, \dots, \frac{a_p}{b_p}$ are the distinct roots

$$m_1, m_2, \dots, m_p$$

$$\text{C.F.} = \phi_1(b_1 y + a_1 x) + \phi_2(b_2 y + a_2 x) + \dots + \phi_p(b_p y + a_p x)$$

Case II

$$D = m = \frac{a}{b} \text{ (m times)}$$

$$\text{C.F} = \phi_1(by + ax) + x\phi_2(by + ax) + \cdots + x^{m-1}\phi_m(by + ax)$$

Example (1)

$$(25D^2 - 40DD' + 16)Z = 0$$

$$D = m, D' = 1$$

$$(5m - 4)^2 = 0$$

$$m = \frac{4}{5}, \frac{4}{5}$$

$$\text{C.F } z = \phi_1(5y + 4x) + x\phi_2(5y + 4x)$$

Example (2)

$$(D^3 - 3D^2D' + 2DD'^2)z = 0$$

$$D = m, D' = 1$$

$$(m^3 - 3m^2 + 2m) = 0$$

$$m = 0, 1, 2$$

$$z = \phi_1(y) + \phi_2(y+x) + \phi_3(y+2x)$$

Particular Integral

$$F(D, D')z = f(x, y)$$

$$z = \frac{1}{F(D, D')} f(x, y)$$

Case1

$$\text{a) } \frac{1}{F(D, D')} e^{ax+by} = \frac{1}{F(a, b)} e^{ax+by} ; F(a, b) \neq 0$$

$$F(a, b) = 0 \text{ විට}$$

$$F(D, D') = \left(D - \frac{a}{b}D'\right)^r G(D, D')$$

$$\text{එවිට } G(D, D') \neq 0 \text{ වේ.}$$

$$\begin{aligned} z &= \frac{1}{F(D, D')} e^{ax+by} = \frac{1}{\left(D - \frac{a}{b}D'\right)^r} \frac{1}{G(D, D')} e^{ax+by} \\ &= \frac{1}{G(a, b)} \frac{1}{\left(D - \frac{a}{b}D'\right)^r} e^{ax+by} = \frac{1}{G(a, b)} \frac{x^r}{r!} e^{ax+by} \end{aligned}$$

Case11

$$\frac{1}{F(D^2, DD', D'^2)} \sin(ax + by) = \frac{1}{F(-a^2, -ab, -b^2)} \sin(ax + by)$$

$$\frac{1}{F(D^2, DD', D'^2)} \cos(ax + by) = \frac{1}{F(-a^2, -ab, -b^2)} \cos(ax + by)$$

$$\ominus \ominus \Im \frac{1}{F(-a^2, -ab, -b^2)} \neq 0$$

Example (1)

$$(D - D')(D - 2D')z = e^{2x+3y}$$

$$\text{C.F} \quad z = \phi_1(y + x) + \phi_2(y + 2x)$$

$$\text{P.I} \quad z = \frac{1}{(D - D')(D - 2D')} e^{2x+3y} = \frac{1}{(-1)(2-6)} e^{2x+3y} = \frac{1}{4} e^{2x+3y}$$

$$\therefore z = \phi_1(y + x) + \phi_2(y + 2x) + \frac{1}{4} e^{2x+3y}$$

Example (2)

$$(D - D')(D - 2D') z = e^{x+y}$$

$$\begin{aligned} \text{P.I} \quad z &= \frac{1}{(D - D')(D - 2D')} e^{x+y} \\ &= \frac{1}{(D - D')(1 - 2 \cdot 1)} e^{x+y} \\ &= \frac{-1}{(D - D')} e^{x+y} \\ &= -x e^{x+y} \end{aligned}$$

Example (3)

$$(D - D')(D - 2D') z = \sin(x - 2y)$$

$$(D^2 - 3DD' + 2D'^2) z = \sin(x - 2y)$$

$$\begin{aligned} \text{P.I} \quad z &= \frac{1}{(D^2 - 3DD' + 2D'^2)} \sin(x - 2y) \quad \text{P.I} \quad z \\ &= \frac{1}{-1^2 - 3[-(1 \cdot (-2))] + 2(-(-2)^2)} \sin(x - 2y) \\ &= \frac{-1}{15} \sin(x - 2y) \end{aligned}$$

Case III

When $f(x, y)$ has the powers of x and y

$$(D^2 - DD' - 6D'^2)z = x + y + 1$$

$$\text{P.I} \quad z = \frac{1}{(D^2 - DD' - 6D'^2)} (x + y + 1)$$

$$= \frac{1}{D^2 \left(1 - \frac{D'}{D} - \frac{6D'^2}{D^2}\right)} (x + y + 1)$$

$$= \frac{1}{D^2} \left[1 + \frac{D'}{D} + \dots\right] (x + y + 1)$$

$$= \frac{1}{D^2} \left[x + y + 1 + \frac{1}{D}\right]$$

$$= \frac{1}{D^2} [x + y + 1 + x]$$

$$= \frac{1}{D^2} [2x + y + 1] = \frac{1}{3}x^3 + \frac{1}{2}x^2y + \frac{1}{2}x^2$$